BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: 021/1

Retyped faithfully from the original board paper — same questions, no changes.

BIOLOGY (PART –II) (OBJECTIVE PART)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 021/1

Note: Note:- Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. (Marks (17)) Each question has four possible answers, Tick (√) the correct answer.

Q1. Condensation of chromosomes reaches to its maximum during;

(A) Diakinesis
(B) Diplotene
(C) Pachytene
(D) Zygotene

Q2. Insulin gene is located on short arm of chromosome;

(A) 7
(B) 9
(C) 11
(D) 19

Q3. EcoR1 is used as;

(A) Vector
(B) Expression system
(C) Restriction enzyme
(D) Gene of interest

Q4. In fish, the gill pouches develop into;

(A) Gills
(B) Pharynx
(C) Eustachian tube
(D) Fins

Q5. All food chains and food webs begin with;

(A) Primary consumers
(B) Secondary consumers
(C) Tertiary consumers
(D) Producers

Q6. In temperate grasslands, the rate of primary productivity is annually;

(A) 4000 g/m²
(B) 700 – 1500 g/m²
(C) 500 – 700 g/m²
(D) 2000 – 2500 g/m²

Q7. The cause of stone cancer is;

(A) Green house effect
(B) Water pollution
(C) Acid rain
(D) Ozone depletion

Q8. Liver synthesizes;

(A) Iron
(B) Glycogen
(C) Glucose
(D) Bile

Q9. In juxtamedullary nephrons, addition capillaries extend down to form;

(A) Peritubular capillaries
(B) Vasa recta
(C) Glomerulus
(D) Loop of Henle

Q10. Vertebral column extends from skull to which region;

(A) Neck
(B) Thorax
(C) Pelvis
(D) Lumber

Q11. The disease which causes the fusion and immobility of vertebral joints is;

(A) Spondolysis
(B) Sciatica
(C) Rickets
(D) Osteomalcia

Q12. Chemical nature of insulin and glucagon is;

(A) Proteins
(B) Amino acid derivatives
(C) Steroids
(D) Polypeptides

Q13. Germinating pollen grain is a rich source of;

(A) Ethene
(B) Gibberellins
(C) Cytokinins
(D) Auxins

Q14. In chick, the egg is fertilized as it passes through its;

(A) Shell gland
(B) Cloaca
(C) Kidney
(D) Liver

Q15. Acetabularia is a/an;

(A) Fungus
(B) Alga
(C) Fern
(D) Gymnosperm

Q16. In bacteria, human and all living organisms, AGA specifies;

(A) Phenylalanine
(B) Leucine
(C) Methionine
(D) Arginine

Q17. A period of extensive metabolic activity in which cell grows in size is;

(A) G_0
(B) G_1
(C) G_2
(D) S

Note: (The End)

BIOLOGY PAPER : PART – II (SUBJECTIVE PART)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 021/1

Note: Note:- Attempt any TWENTY TWO (22) short questions in all selecting eight from Q. 2 and Q. 3 each and six from Q. 4. (22 x 2 = 44)

Note: SECTION – I

Q2. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(1) What do you mean by Homeostasis?

(2) What are xerophytes? Write their adaptations.

(3) Differentiate between Osmoconformers and Osmoregulators.

(4) What are collenchyma cells?

(5) What is ribcage?

(6) Name the types of cells associated with bones.

(7) What is menupause? At what age is starts?

(8) Define parthenocarpy with examples.

(9) What are planktons? Give their types.

(10) What do you mean by taiga?

(11) What is ozone layer? Give its advantage.

(12) What is acid rain?

Q3. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(1) What are the commercial applications of Auxin?

(2) What is Neuroglia? Give its role.

(3) Name the Hormones secreted by pancrease and their role.

(4) What is Gene linkage?

(5) Define Gene and Locus.

(6) What are Sex-linked traits?

(7) Compare Ex-vivo gene therapy with in-vivo gene therapy.

(8) What are three possible ways to get a gene of interest for gene cloning?

(9) Discuss Sanger's Method of Gene sequencing.

(10) Define ecosystem.

(11) Write down the significance of Root Nodules in plants.

(12) Define parasitism and give one example.

Q4. (Marks (2 x 6 = 12)) Write short answers of any six questions.

(1) What are lateral meristems?

(2) Explain regeneration in Salamander.

(3) Name four types of Chromosomes.

(4) What are Okazaki fragments?

(5) What are non sense codons?

(6) Write a note on Diakinesis.

(7) What is Turner's Syndrome?

(8) What is Genetic drift?

(9) Explain Endosymbiont Hypothesis.

> Note: SECTION – II

> Note: Note:- Attempt any three questions. (3 x 8 = 24)

Q5. (Marks (08)) (a) Describe the role of liver as a major homeostatic organ. (04)

Q. (Marks (04)) (b) Describe parasitic and mutualistic relationships in an ecosystem.

Q6. (Marks (08)) (a) Define Antagonism. Discuss the phenomenon with the example of elbow joint. (04)

Q. (Marks (04)) (b) Define transcription. Discuss the detailed steps in prokaryotes.

Q7. (Marks (08)) (a) Explain Feedback Mechanism. (04)

Q. (Marks (04)) (b) Write note on importance of Forests.

Q8. (Marks (08)) (a) Explain about "Seed Dormancy". (04)

Q. (Marks (04)) (b) Define and explain Mendel's Law of Segregation.

Q9. (Marks (08)) (a) What is regeneration? Explain it with the help of examples in different groups of animals. (04)

Q. (Marks (04)) (b) Describe Biogeography and molecular biology as an evidence of evolution.

> Note: (The End)